3GPP Technical Specifications: 5G NR Support for Unmanned Aerial Vehicles (Releases 15-18)

3rd Generation Partnership Project (3GPP TS 22.125 / TS 23.256)

Regulatory Domain: Global Cellular Telecom Standards for UAS

TS 22.125 - UAV Communication Service Requirements

Defines communication KPIs for UAS Command and Control (C2): ultra-reliable low latency (URLLC), packet error rates under 10^-4, and seamless cell handover to prevent line-of-sight dropouts in urban canyons.

TS 23.256 - Architecture Enhancements for 5G UAS

Specifies aerial beamforming techniques, altitude-dependent interference mitigation, and dynamic QoS flow allocation for priority unmanned aerial missions.